Space Station Safety Equipment Detector

An AI-Powered Application for Detecting Critical Safety Equipment

Key Technologies

YOLOv8 Object Detection • Falcon Digital Twin Platform
Streamlit Web Interface • Real-time Processing

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Contents

1 Overview

This document presents a comprehensive AI-powered application designed for detecting critical safety equipment in space station environments. The system leverages YOLOv8 object detection technology, trained on synthetic data generated from the Falcon digital twin platform, to identify seven types of essential safety equipment with high accuracy and reliability.

2 Features

The application provides a robust suite of capabilities designed for operational excellence:

- **Real-time Detection**: Instantaneous identification of 7 types of safety equipment
- **High Accuracy**: Advanced model trained on high-quality synthetic datasets
- **User-Friendly Interface**: Dual interface options (Web-based Streamlit and Desktop Tkinter)
- **Confidence Adjustments**: Configurable detection thresholds for optimal performance
- **Detailed Reports**: Comprehensive per-object statistics and visualizations
- **Continuous Updates**: Seamless integration with Falcon platform for ongoing improvements

3 Detected Equipment Categories

The system is capable of identifying the following critical safety equipment:

1. Fire Extinguisher
2. Fire Alarm
3. First Aid Box
4. Oxygen Tank
5. Nitrogen Tank
6. Safety Switch Panel
7. Emergency Phone

4 System Requirements

4.1 Hardware and Software Specifications

5 Usage Instructions

5.1 Streamlit Web Application (Recommended)

The Streamlit interface provides the most comprehensive feature set and is recommended for most users.

Component	Requirement
Operating System	macOS, Linux, or Windows
Python Version	3.8–3.11
Memory (RAM)	8GB minimum (16GB recommended)
Storage	2GB free space
GPU Support	Apple Silicon (M1/M2/M3) via MPS

Table 1: System Requirements

```
# Start the application
streamlit run apps/streamlit/streamlit_app.py

# Application opens automatically in browser
# Default URL: http://localhost:8501
```

Listing 1: Launching Streamlit Application

Key Features:

- Upload images or video files
- Real-time confidence and IoU threshold adjustments
- Comprehensive detection statistics
- Download annotated results

5.2 Desktop GUI Application

For users preferring a standalone desktop application:

```
# Run the desktop application
python apps/desktop/tkinter_app.py
```

Listing 2: Launching Desktop Application

Key Features:

- Integrated file browser
- Adjustable detection parameters
- Real-time results display
- Lightweight and responsive

6 Model Performance

6.1 Overall Metrics

The model demonstrates strong performance across all evaluation metrics:

Metric	Value
mAP@0.5	76.6%
mAP@0.5–0.95	63.4%
Inference Speed (CPU)	238.7ms
Training Images	2,103 (1,767 train + 336 val)
Architecture	YOLOv8s (Small)

Table 2: Overall Model Performance

Object Class	Precision	Recall	mAP@0.5
Fire Extinguisher	93.2%	56.8%	71.2%
Fire Alarm	93.5%	76.1%	86.0%
First Aid Box	87.9%	68.9%	82.6%
Oxygen Tank	95.7%	76.4%	86.1%
Nitrogen Tank	88.3%	69.3%	81.0%
Safety Switch Panel	91.3%	55.0%	68.2%
Emergency Phone	91.4%	52.2%	61.1%

Table 3: Performance Metrics by Equipment Class

6.2 Per-Class Performance Analysis

7 Falcon Integration Pipeline

7.1 Monthly Update Workflow

The system implements a structured monthly update cycle to maintain and improve model performance:

7.1.1 Step 1: Performance Monitoring

```
python monitor_production.py --logs logs/month_1/
```

Listing 3: Production Performance Analysis

7.1.2 Step 2: Scenario Generation

- Identify areas of weak detection performance
- Design targeted scenarios (lighting conditions, viewing angles, occlusions)
- Generate 500–1000 new labeled images via Falcon platform

7.1.3 Step 3: Dataset Integration

```
python merge_datasets.py \
    --original datasets/v1.0 \
    --new falcon_generated/month_2 \
    --output datasets/v1.1
```

Listing 4: Merging Datasets

7.1.4 Step 4: Model Retraining

```
python train.py \
    --data datasets/v1.1/config.yaml \
    --epochs 50 \
    --weights models/v1.0/best.pt \
    --name v1.1
```

Listing 5: Training Updated Model

7.1.5 Step 5: Validation and Deployment

```
# Compare performance between versions
python compare_models.py --model1 v1.0 --model2 v1.1

# Deploy improved model
cp runs/train/v1.1/weights/best.pt \
   ./src/models/safety_equipment_model.pt
```

Listing 6: Model Comparison and Deployment

7.2 Falcon Platform Advantages

- **Unlimited Data Generation**: On-demand synthetic image creation
- **Perfect Annotations**: Automatic, error-free labeling
- **Scenario Control**: Precise manipulation of environmental variables
- **Cost Effectiveness**: Elimination of manual labeling expenses
- **Rapid Iteration**: Development cycles reduced from weeks to hours

8 Optimization Guidelines

8.1 Detection Parameter Tuning

Use Case	Confidence	IoU
High Precision	0.4	0.5
High Recall	0.2	0.4
Balanced (Default)	0.25	0.45

Table 4: Recommended Parameter Settings

8.2 Image Quality Requirements

For optimal detection performance, ensure:

- Adequate lighting conditions (neither too dark nor overexposed)
- Sharp, focused imagery
- Object dimensions exceeding 20 pixels
- Minimal motion blur

8.3 Troubleshooting Common Issues

Issue	Solution
No detections	Decrease confidence threshold
Excessive false positives	Increase confidence threshold
Overlapping bounding boxes	Adjust IoU threshold
Slow inference	Use YOLOv8n or reduce image resolution

Table 5: Common Issues and Solutions

9 Hackathon Compliance

This application fulfills all bonus challenge requirements:

9.1 Application Components

- Fully functional detection interface (web and desktop)
- Integrated model inference system
- Intuitive user controls
- Comprehensive results visualization

9.2 Falcon Integration Strategy

- Structured continuous learning pipeline
- Defined monthly update schedule
- Systematic scenario generation methodology
- Robust performance monitoring framework

9.3 Documentation Package

- Complete usage instructions
- Detailed installation procedures
- Update and maintenance protocols
- Comprehensive technical specifications

10 Conclusion

The Space Station Safety Equipment Detector represents a state-of-the-art solution for automated safety equipment monitoring in space station environments. By leveraging advanced AI technologies and maintaining a continuous improvement cycle through Falcon platform integration, the system ensures reliable, accurate detection capabilities that evolve with operational requirements.

For Additional Information:

Refer to `falcon_integration_plan.md` for detailed implementation guidance and technical specifications.